Caring about Care: A Meta-Narrative Review of HCI Research on Care

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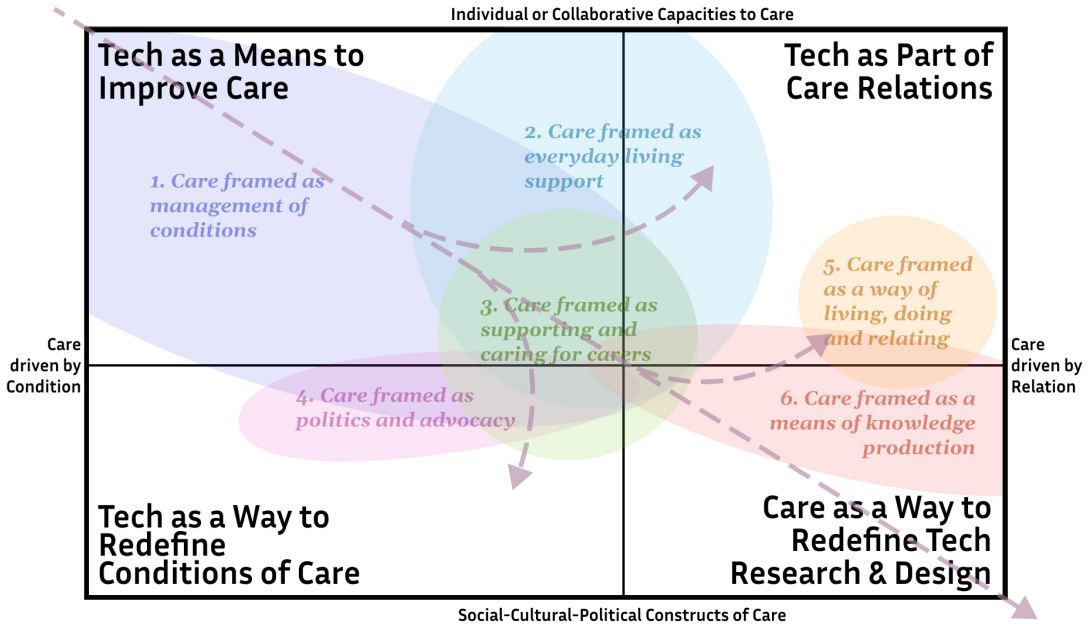


Figure 1: In this paper, we present a conceptual map connecting and positioning the six framings of care identified from our review, along with their representative care-tech relations, and the temporal development of the field. The underlying papers and their metadata are available for interactive exploration on our website: https://care-hci.org/.

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Abstract

The number of HCI papers on care has grown rapidly in recent years. Despite growing interest in care both as an application domain for technology and as an ethical stance in research and design, our integrated understanding of the concept is limited. It remains unclear how various application areas of care relate to one another,

to what extent their underlying assumptions align or contradict, and how they collectively shape HCI discourse on care. To address this, we present a meta-narrative review of 317 SIGCHI papers on care. We first outline the landscape of care in HCI. We then present six paradigmatic framings of care, and a conceptual map that positions these framings in relation to each other, their representative care–tech relations, and the temporal development of the field. We conclude by discussing the implications from the review, as well as gaps in the field and future directions.

CCS Concepts

• **General and reference → Surveys and overviews**; • **Human-centered computing → HCI theory, concepts and models**; *Interaction design theory, concepts and paradigms.*

Keywords

Care, Literature Review, Feminist Ethics of Care, Matters of Care

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1 Introduction

The field of Human-Computer Interaction (HCI) has seen growing interest in researching and designing for and with care. Though historically centred around health and medical notions of care, recent years have seen a shift towards a broader and relational understanding of care, drawing on feminist care ethics and more-than-human theories [133, 193]. This development from viewing *care* as only a setting for research or application domain for technology, to *caring* as an epistemological orientation and ethical stance for research and design, has expanded how care is approached in HCI [93, 127, 172, 191]. However, attempts to embrace the complexity of care frequently face difficulties in reconciling diverse perspectives and approaches. As Helms and Fernaeus [93] observed, *"care means and does many things, and within its ambivalence of significance and ontology are tensions and contradictions"*. The slippery, elusive, and multivocal nature of care in everyday and academic use can make it—as an underpinning concept for design—challenging to apply and make explicit to others [135]. Given the expanding body of research engaging with care, there is a need to examine and elucidate its role and significance within HCI.

Within the field, there have been reviews summarising studies of care in particular domains (detailed in Section 2). Yet, it remains unclear how these strands relate to one another, how they collectively shape HCI discourse on care, and to what extent their underlying assumptions align or contradict. Therefore, this paper reviews HCI research on care over time, examining how the concept is framed, and proposes a conceptual map to help researchers navigate, position, and articulate care in their work. The aim of the paper is not to reach a unifying definition of care, but to learn across domains, methods, and theoretical perspectives in order to provide clarity in meaning to support its *"transformative potential"* [134].

This paper offers three contributions. First, we present an overview of 317 SIGCHI publications on care. We highlight application areas of care that HCI research has been engaged with, detailing who, what, and where has been included or overlooked. Second, we define six paradigmatic framings of care from our analysis—care framed: as *management of conditions*; as *everyday living support*; as *supporting and caring for carers*; as *politics and advocacy*; as a *way of living, doing and relating*; and as a *means of knowledge production*. Third, we introduce a conceptual map along with an interactive website for care in HCI, from which we discuss connections and temporal development across the framings, including reflections on tensions and future research directions.

2 Related Work & Research Questions

As this paper is a review of care research in HCI, we are not going to present a typical related work of research on or about care. Instead, here we summarise previous reviews that have addressed different aspects of care in HCI. We then position the scoping of our research questions and situate our research interests and intended audience in relation to this prior work.

2.1 Literature Reviews on Care in HCI

There have been a number of efforts to start disentangling research on care in HCI through reviewing research published on the topic. An early effort is Fitzpatrick and Ellingsen's [70] 2013 review on CSCW research in healthcare. They emphasised how care work is collaboratively practised and technologically supported. Their review revealed a rich conceptual understanding of the daily work of healthcare staff and foregrounded a gap in understanding of healthcare practices at a policy level. In 2015, Nunes et al. [149] reviewed self-care technologies in HCI that support patients (and their carers) with chronic conditions. Their review was motivated by a concern about the dominant medical perspective influencing the design of self-care technologies, and identified opportunities to include people's everyday care experiences and their relationships with informal carers. Regarding the aims of this paper, the authors call for future reviews that include cross-disciplinary perspectives and a diversity of application areas to expand understanding of care and its impact on technology design. Later reviews have applied a conceptual lens of care to investigate a particular domain or application area. This includes a 2022 review [182] of technologies for children with ADHD that was grounded in understanding *care ecosystems* and a 2023 review [212] that used a *care ethics* perspective to investigate social robots in aged care.

Most recently, and of most direct relevance to the aims of our review, is the work of Petterson et al. [158] published during the writing of this paper. Their study is motivated by compounding ethical concerns in HCI, including the environmental crisis and systemic social inequities. This grounds their investigation on how care, and in particular feminist ethics of care, offers a critical lens for reimagining sustainability in relation to sociotechnical systems. They reviewed 93 papers that engaged with "networked (as opposed to between two individuals) understandings of care", and identified six ways in which care is conceptualised. They further suggest ways for HCI researchers to engage with critical theories of care.

Similarly to Petterson et al. [158], we are interested in mapping HCI research on care to understand its breadth, gaps, and possibilities. However, our review differs in several important ways. Foremost, Petterson et al. did not seek to provide a comprehensive review and were intentionally selective in terms of inclusion, limited to a ten-year period. Our review is guided by an interest in developing a pluralistic understanding of care research in HCI, including how framings of care have developed and changed over time. Second, Petterson et al. [158] depart from a feminist ethics of care perspective to investigate how this relates to particular design challenges (i.e. *"care offer[ing] a critical lens for re-imagining sustainability"* [158]). While we orient our own interdependent research agendas within feminist ethics of care, it is not the theoretical motivation for our review or the generative foundation of our contribution. Instead, our review focuses on how HCI researchers with diverse aims and interests in care—such as an application area and/or a specific theoretical framing—can better situate, relate to, and build upon one another. However, our findings are complementary to those of Petterson et al.'s more specific review, and we return to discuss this in Section 7.1.

2.2 Scoping of Research Questions

Our review aims to clarify how care is applied, framed, and conceptualised in HCI. It is based on three research questions:

- **RQ1.** What application areas of care have HCI research been engaged with?
- **RQ2.** What paradigmatic framings of care have been adopted, and what are their underlying assumptions?
- **RQ3.** How have different paradigms of care in HCI developed over time, and what transformative potentials of care for HCI does this reveal?

RQ1 seeks to identify specific settings, contexts, and actors engaged with by HCI research on care, and the methods used to approach these. RQ2 seeks to understand the epistemological orientations on care and how they have broadened from a condition or setting to a way of approaching other settings, contexts, and uses of technology. RQ3 seeks to explore how HCI's engagement with care relates to the temporal development of the field, as seen through care-tech relations and waves of HCI. This grounds implications for technology design and research from our review.

2.3 Author Positionality & Intended Audience

The authors of this paper work across diverse care domains—including healthcare, aged care, intimate care, and care ethics—and share an interest in feminist care theories, albeit with varying expertise. Our motivations and understandings of care are shaped by our professional expertise, cultural backgrounds, and personal identities and experiences. All but the last author identify as female. We come from the UK, China, Australia, and North America, and each bring our cultural understandings and personal experiences of care (e.g., as care caregivers for elder relatives). Despite having an understanding of theories around care and care ethics, many of us find it challenging to translate into practice, and communicate the meaning of "broader understanding of care" when publishing in our respective fields. Through this review, we aim to bridge care theories with practical contexts, and offer suggestions for future research to clarify and make explicit its pragmatic intentions and theoretical agendas.

We envision three main types of audience:

- researchers who are new to the field, to help them build an overall understanding of the landscape of care;
- researchers from a care ethics background, to highlight the broad application domains of care in HCI and pragmatic issues where care can be applied;
- researchers working in a specific care domain, to encourage them to explore related fields and alternative ways of conceptualising and operationalising care in research.

3 The Review Approach

We adopted the meta-narrative review approach proposed by Greenhalgh et al. [81], which is well-suited for synthesising large, complex bodies of literature. This approach is particularly valuable for examining heterogeneous literature, where researchers have conceptualised and investigated similar concepts in different ways. Given that HCI is an interdisciplinary field drawing on theories, methods, and paradigms from multiple disciplines, meta-narrative review is appropriate here. Our review followed its six guiding principles: pragmatism, pluralism, historicity, contestation, reflexivity, and peer review (see Figure 2) [81].

Principle	Description
1. PRAGMATISM	What to include is not self-evident. The reviewer must be guided by what will be most useful to the intended audience(s), for example, what is likely to promote sense making.
2. PLURALISM	Topic should be illuminated from multiple angles and perspectives, using the established quality criteria appropriate to each. For example, reviewers should avoid beginning with a single 'preferred' perspective or methodological hierarchy and proceed to judge work in other traditions using these external benchmarks.
3. HISTORICITY	Research traditions are often best described as they unfolded over time, highlighting significant individual scientists, events and discoveries which shaped the tradition.
4. CONTESTATION	Conflicting data from different research traditions should be examined to generate higher-order insights. For example, about how different research teams framed the issue differently or made different assumptions about the nature of reality.
5. REFLEXIVITY	Reviewers must continually reflect, individually and as a team, on emerging findings.
6. PEER REVIEW	Emerging findings should ideally be presented to an external audience and their feedback used to guide further reflection and analysis.

Figure 2: Six principles of meta-narrative review.

3.1 Data Collection

We focused on peer-reviewed literature in ACM SIGCHI conferences, the main publication venues for the HCI community. This ensured a manageable yet representative corpus for examining how HCI has conceptualised and operationalised care. After testing various options, we conducted a systematic search using only the terms "care" and "caring" in titles, abstracts, and keywords. This allowed us to examine the diverse ways the concept is framed, without being restricted by predefined terminologies. As care is often used with modifiers—such as "self-care", "care ethics", "networks of care", "caring through data"—searching the root terms captured all these

variations. No start date was applied, allowing us to trace developments over time, while the end date was set to 31st December 2024 to ensure equal representation of annual conference proceedings. The search yielded 1187 papers, from which 418 duplicates were removed, resulting in 769 papers for screening.

3.2 Screening Criteria

Since the study aims to map the breadth of HCI research on care, the only inclusion/exclusion criteria we applied is whether the paper has a focus or contains sufficient discussion on care for analysis. This was determined by conducting a full text search for "care" and "caring" (including all their derivative forms) in the paper and reading through sections where they appear. The 769 papers were distributed among three of the authors (called "reviewer" in the following text), and a spreadsheet was used to record decisions and reasoning. We excluded publications if:

- "care" appeared only in search fields (titles, abstracts, keywords), session titles, or references, but not in the main text;
- "care" served solely as an abbreviation for something else;
- "care" was mentioned only in background sections or future plans whilst the main study focused elsewhere;
- "care" was only stated incidentally, rather than as an intentional application domain or concept (e.g., "care must be taken", "carefully choose");
- "care" functioned solely as a proper noun (e.g., use "caregivers" to refer to participants) without any other discussion of care in the study;
- there were insufficient mentions or discussion of care to provide adequate material for analysis.

For borderline cases, inclusion depended on the paper's relevance to the review (i.e., whether it contained at least one paragraph explicitly discussing "care" and contributing new insights or perspectives on care in HCI). Where uncertain, the reviewer marked the paper and indicated a leaning towards inclusion or exclusion. All unsure papers underwent a second review, and reviewers discussed until consensus was reached.

We set the filter to "research articles" (full papers) during search, however some retrieved items were workshop proposals, extended abstracts, or talk biographies. These were excluded as they had not typically undergone full peer review nor contained complete research. We also excluded one paper without an accessible full-text [124] and four literature reviews [65, 116, 179, 182]. The final corpus comprised 317 papers. Figure 3 illustrates the search and screening process.

3.3 Analysis Approach & Process

We adopted an interpretive approach in analysis, as the review aims to be formative (i.e., shaping, guiding, and being generative for future research) rather than summative (i.e., judging the research outputs). Following Greenhalgh et al. [81], we acknowledge that constructing narrative threads from a heterogeneous body of literature inevitably involves subjective and negotiable choices. To mitigate potential bias, we involved multiple authors and external experts in the process. Their areas of expertise and roles in the review are summarised in Table 1. An overview of our analysis process is in Figure 4.

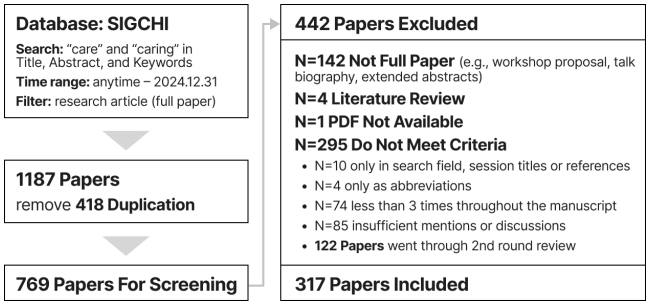


Figure 3: Overview of the search and screening process.

Table 1: People involved in the review and their area of expertise. (*Note: author 5 and 6 were not included here as their expertise lies in the data visualisation tool.*)

	Professional Area	Involved In
A1	Ethics of Care, Aged Care	Coding, synthesis discussion, and finalising of the codes & results
A2	Healthcare, Service Design	Coding healthcare thread of papers
A3	Aged Care, Ethics	Coding more-than-human care thread of papers
A4	Aged Care, Speculative Design	Coding aged and dementia care thread of papers
A7	Intimate Care, Ethics of Care	Synthesis discussion on the findings
A8	Self-care, Feminist HCI	Synthesis workshop on self-care
A9	Aged Care, Participatory Design	Double review papers & synthesis discussion
E1	Ethics of Care, Community-based Design	Synthesis workshop on care in design & research
E2	Ethics of Care, Mental Healthcare	Synthesis workshop on care in design & research

Pilot Coding. Guided by our research questions, we conducted both semantic coding (explicit content such as care settings for **RQ1**) and latent coding (underlying assumptions about care and technology for **RQ2**) of thematic analysis [28]. First, five of the authors conducted a full-day pilot analysis, structured into four sessions of individual coding and group discussion. This process was guided by three broad sensitising questions (see Figure 4-1). The aim was to exchange perspectives and align analytical goals for individual coding. The pilot analysis was repeated several times, moving from printed papers to ATLAS.ti software.

Coding Schemes. Papers were then distributed among the first four authors for individual coding (called "coder" in the following text), with weekly meetings to review codes. A coding template

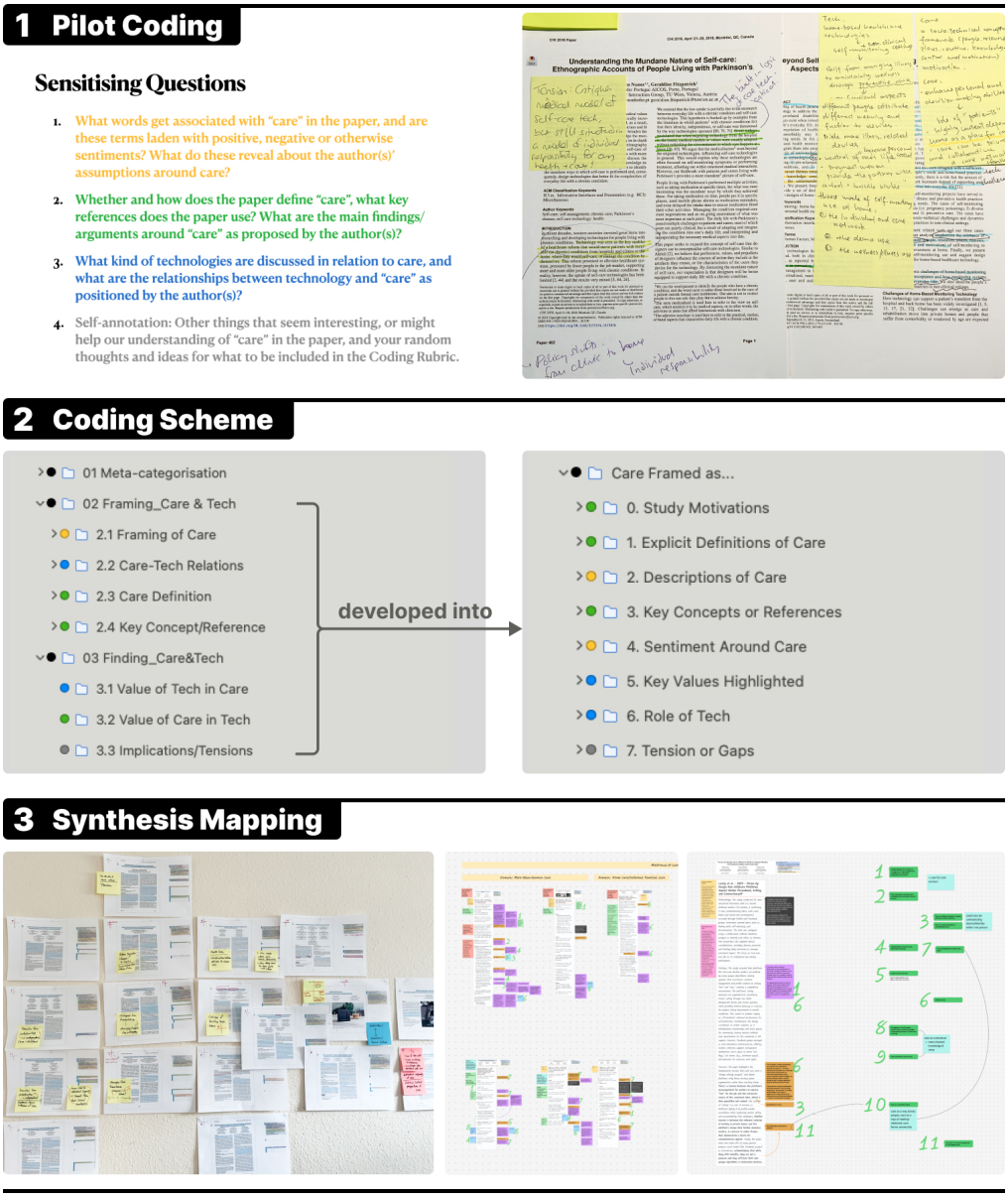


Figure 4: Snapshots of our three-step analysis process.

based on the pilot coding was provided, but codes were generated inductively by each coder. Coders first recorded the meta-data of each paper, then searched for "care" and "caring" in the full text, reading the surrounding context, annotating relevant sentences, and adding a code summarising that relevance. The initial coding scheme comprised three large categories. The first was *"Meta-categorisations"*, which includes the study domain, settings, actors in care, data collection and analysis methods, positionality statements, and the paper's contribution type. The second was *"Framings of Care & Tech"*, which included any explicit definitions of care, key concepts or citations, assumptions about care, and relations between framings of care and the role of technology. The third category, *"Findings about Care & Tech"*, had no predefined sub-categories but captured any other discussions about technology design in the paper's findings.

These categorisations were constantly revisited and reorganised during the coding process to identify broader themes. For example, category 2 and 3 were separated into "framings of care" and "care-tech relations", with other codes re-grouped under them. We also added analysis of study motivations, as we found it relevant to understanding the paper's framing (e.g., why study care, why technology is relevant). The final coding scheme is in Appendix A.

Synthesising & Mapping. After initial coding, the team reconvened to merge codes, compare findings, and identify overarching trends, themes and tensions across the dataset (**RQ3**). Then, the

first author reviewed all papers to further organise the codes and summarise insights. During this process, discussion workshops were held with the wider author team and external care experts. The aim was to seek input on the analysis from their specific care expertise, and explore organising dimensions or potential mappings (see Figure 4-3). This process was also supported by the visualisation tool (see next subsection). Drawing on these insights, the first author formed an initial categorisation of the framings and their definitions. She then re-reviewed all papers referring to these framings, while the last author double-reviewed about one third of the papers to further refine the scope of the framings. This resulted in the final six paradigmatic framings of care in Section 5.

We recognise that the formation of the framings and unfolding storylines of a large heterogeneous body of literature is not straightforward, but an iterative, messy, sense-making process. However, as argued by [81], this process is necessary and important for the synthesis of complex evidences. We acknowledge that the framings and mappings present are our interpretations of the claims in the literature. They represent one way of organising the literature and their storylines around care in HCI, but are not the only way.

3.4 Development of a Data Visualisation Tool

To trace the historical roots and temporal development of the papers, three of the authors drew upon their prior experience in bibliometrics analysis and data visualisation to extract reference data, examine co-citation relationships, and map manually labelled data. This began as an informal exploration to support our data analysis, but we subsequently recognised its potential as both a navigation tool for readers and an instantiation of the paper's argument. Interactive maps, extending linear descriptions, can better demonstrate the nuanced relationships between papers and across domains. While our findings were not organised according to the website, it remained an important reference throughout the analysis and writing. Readers can access and interact with the tool here: **https://care-hci.org**. The website provides the full list of included papers with metadata and search and filter functions. It also serves as a way for us to maintain the review and receive reader's feedback, such as suggesting corrections or requesting the addition of a missing paper after publication.

4 Landscape of Care in HCI

In this section, we describe the overall characteristics of the included papers, their application domains and scope, and the study methods and practices around positionality.

4.1 Characteristics of the Included Papers

Our review included 317 papers on care published between 2008 and 2024 across 18 publication venues. The number of papers on care published annually has increased substantially, from fewer than five average per year in the first 5 years to 60 in 2024 (see Figure 5). The majority of the papers in our corpus were published in CHI (n=215, 68%), followed by DIS (n=42, 13%), and CSCW (n=29, 9%), which together account for 90% of papers (see Figure 6).

We refer to Wobbrock and Kientz's [208] classification to understand the papers' contribution types. We classified publications into

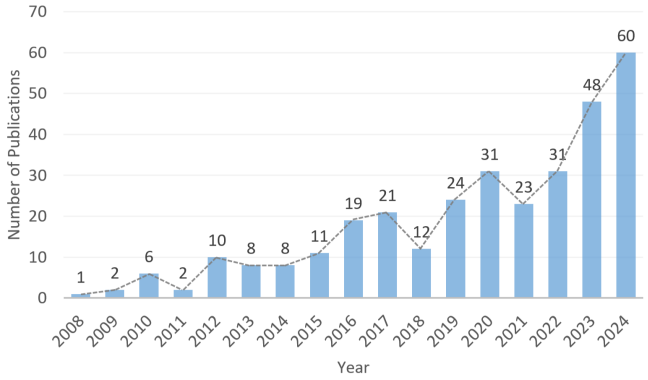


Figure 5: Number of publications across years.

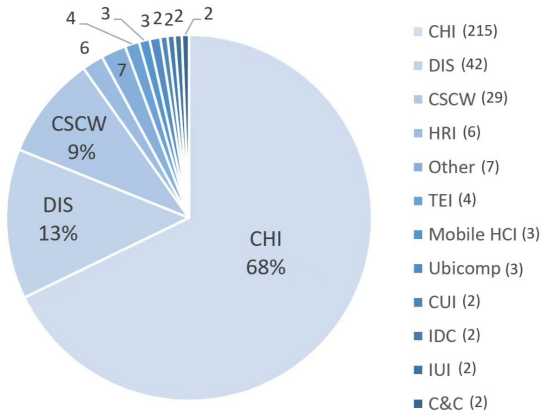


Figure 6: Distribution of publication venues.

(1) empirical studies about people (n=107, 34%), (2) empirical studies about how people use a system (n=121, 38%) (3) artifact or system (n=48, 15%), (4) methodological (n=31, 10%), (5) theoretical or opinion (n=10, 3%).[1] Overall, HCI research on care has been dominated by empirical studies to understand people and their interactions with systems, while theoretical and methodological contributions have been limited. It is also worth noting that all theoretical papers appeared after 2019.

4.2 Application Domains and Scope

Through the review, we saw a diverse range of application domains of care, spanning from formal healthcare, paid and unpaid care at home, care facilities and organisations, self-care, social and communal care, more-than-human care, to care applied in design and research process. We present the *actors*, *settings*, *regions*, and *technologies* involved in care in the included papers below. The aim is to illustrate the breadth of topics related to care in HCI and highlight what has been included, and what may have been missed.

[1] Many papers in artifact, methodological, theoretical contributions also include an empirical study element, we categorised them based on the primary contribution claimed by the authors.

Actors in Care. Of the papers reviewed, 43% (*n*=135) involved people with health conditions, mainly chronic ones (e.g., cancer (*n*=5), HIV (*n*=4), diabetes (*n*=4), and Parkinson's (*n*=3)), or referred more generally to "patients". The actors involved also demonstrate care as a topic of importance across the life course, though research attention is uneven. Only a few papers examined care relating to children (*n*=16)) and adolescents (*n*=6)), and mostly those with health conditions or disabilities. In contrast, much greater emphasis was placed on older adults (*n*=37), regardless of health status, and on people with dementia specifically (*n*=46).

Women's health also received some attention (*n*=16), centred on pregnancy (*n*=9) and intimate care needs (*n*=7). A small number of papers looked at underserved or marginalised populations, including rural communities (*n*=4), Black Americans (*n*=4), and one study on trans and non-binary people [206]. A further subset examined groups requiring particular social or care support, such as refugees [25, 188], incarcerated individuals [77], veterans with PTSD [66], and survivors of intimate partner violence [195].

Caregivers were also involved alongside people receiving care (*n*=176), though only a small subset focused on caring for them. Professional care workers, including clinicians, nurses, and care home staff were the most common (*n*=118), followed by unpaid carers and family members (*n*=68). Formal home care workers have been studied after 2020, and the research to date covered only a few roles in the USA (i.e., home health aides, childcare worker, and house-cleaners). 8 papers highlighted volunteers as key actors in care, while 12 studied parents as primary caregivers of children. General tech users without specific care-related needs were more included in research after 2020 (*n*=39). Overall, HCI research on care is dominated by a focus on chronic illness and older adults, with many other groups receiving limited attention.

Settings of Care. Besides 85 papers not specifying settings, the most common ones were clinic or hospital (*n*=77, 24%), home (*n*=70, 22%), and care facilities (care homes, nursing homes, and rehabilitation facilities) (*n*=48, 15%). 21 papers examined care in online or gaming environments, and 29 looked at community contexts, including churches and maker communities. Only 3 papers addressed care in public spaces such as playgrounds and cafés [67, 94, 143], and one paper each on diabetes self-care at university [80], intimate care in the workplace [209], and caring for microbes in laboratories [37]. Overall, HCI research on care largely focused on formal care settings and private homes, overlooking critical care needs—such as disease management or breastfeeding—in public, non-clinical spaces where care is not the primary activity.

Study Regions. Most studies happened in North America (*n*=139, 44%), dominated by the USA (*n*=125) and Canada (*n*=12). Europe (*n*=114, 36%) formed the second-largest cluster, with the majority from the UK (*n*=54), where a large proportion focused on dementia care. Asia (*n*=27, 9%) was represented by India (*n*=13), Japan (*n*=6), Singapore (*n*=4), China (*n*=3), and Korea (*n*=2). Oceania (*n*=11) was limited to Australia. 4 studies were conducted in Latin America, including Chile [84, 85] and Mexico [49, 206], most focused on aged care. Besides that, there are three studies in Africa [68, 138, 153], two of them on HIV care. Overall, the distribution shows a strong dominance of research in North America and Western/Northern Europe, with far fewer studies from the rest of the world.[2] Notably, only three papers referring to feminist care ethics were conducted outside North America, Europe, or Australia.

Technologies in Care. Most technologies studied were screen-based applications: 23 well-established tools (e.g., social media, instant messaging, video calls, and other publicly accessible apps or websites) and 83 bespoke applications. Most bespoke apps were healthcare-related, including clinical decision and communication support platforms (*n*=37) and mHealth applications (*n*=18). 37 papers explored self-made tangible artefacts, such as digital jewellery, more-than-human living artefacts, and customed wearables. 138 papers discussed general technological implications. This includes tracking or monitoring technologies (*n*=32), IoT, smart homes, or ambient systems (*n*=13), AI or generative AI (*n*=20), augmented or virtual reality (*n*=9), robots or chatbots (*n*=28). Many of these technologies have been studied more over the past five years, with AI seeing a particular boom.

4.3 Study Methods and Positionality

We categorised the data collection and analysis methods and noted positionality statements in the included papers. The most common methods were *interviews* (*n*=222) and *observational or ethnographic work* (*n*=105), indicating that many papers focused on understanding people and their contexts. *Co-design and workshop* methods became more common around 2016 and popularised in the past two years. *Research-through-design* and *first-person or autobiographical* approaches emerged in the corpus around 2018 and have also seen greater uptake recently.

Data Analysis. Most studies used qualitative data analysis. *Thematic analysis* (*n*=178) and *grounded theory* (*n*=51) together accounted for 72% of the papers. We noted **visual methods** (*n*=20)—such as affinity diagramming, visual mapping, and zine-making—as a distinctive approach used by design researchers, often without explicit methodological reference, and sometimes combined with thematic analysis to map data and generate insights. Given the homogeneity of approaches, we looked at the less common methods used only once or twice as potential alternatives for future research. While thematic analysis and grounded theory focus on identifying themes across a dataset, **explication de texte** (used in [7]) enables in-depth engagement with a single piece of content. **Feminist content analysis** supports examining the ideas, power structures and norms reproduced in data, and was used by Spors et al. to analyse the ideologies embedded in commercial self-care apps [180]. **Diffractive analysis** intentionally looks for and values differences in data. Akbaba et al. provided an illuminating account of their method choice, from open coding to deliberately seeking out differences, linking this to their positionality and focus on matters of care [2].

Practice of Positionality. Explicitly stating authors' positionality is a relatively recent practice and appeared in a small proportion of the papers (*n*=66, 21%). The earliest examples in our dataset were a 2017 paper on dementia care [144] and two 2019 papers that discussed care in design research [98, 167]. Around 2021, positionality

[2] 20 studies spanned multiple countries, and 5 were mixed or unspecified.

statements became a more common practice spanning all domains of care.[3] Drawing on a recent review on positionality practices in HCI [175], we categorised these reflections into three types.

The first is where authors discuss how their previous experience and relationship with the **studied population** influenced recruitment and engagement. For instance, Brozena et al. reflected on how lived experience with bipolar disorder helped them handle potentially sensitive topics [29], and Wani et al. discussed how growing up in the study area and speaking the local language supported their access to, and understanding of, participants [202]. Meanwhile, researchers also reflected on the limitations of accessing participants via these channels [112], and the challenges of not belonging to the same social class or group as participants [6, 111].

The second is reflections around **study methods**, where authors disclose their positionality (e.g., professional interests and attitudes towards the topic) as part of a constructionist approach to interpretation [54, 57, 180, 211]. For example, Karusala et al. noted that the author's research and personal experiences have sensitised her to *"care work as a site for understanding power dynamics around gender and capital, shaping her approach to data analysis"* [112]. However, we also identified some seemingly *"performative" positionality statements* (as called in [175]), where authors' backgrounds and involvement in the study were disclosed, but their impact on the research was not discussed (e.g., [97]).

The third is where authors are transparent about how their professional and personal experiences have shaped the **study focus** (e.g., [209]). While some papers described efforts to prevent authors' own beliefs outweighing participants' views [45], others embraced their subjectivity, treating it as a valuable source of *"partial, situated and contextual"* knowledge [79]. This is particularly evident in papers using autoethnographic or autobiographical approaches, where the author's positionality often influenced every aspect of the study and reflexivity was woven throughout the paper (e.g., [20, 75, 76, 93, 195]).

In addition, **"care"** was also mentioned in positionality statements. In some cases, it appeared as everyday language (e.g.,*"if the study is conducted collaboratively and with care"* [185], *"commit to a careful and contextual interpretation"* [45]). In others, care was discussed as a way of navigating positionality. Howard and Irani, for example, described how care shapes ethnographic work and the way of knowing when participants care deeply about the topic [98]. Meanwhile, Garrett et al. called for critical attentiveness to our positionalities and values, and for making visible our ethical positions as a matter of care in research [76].

5 Paradigmatic Framings of Care

In what follows, we present six paradigmatic framings of care. The presentation of the framings draws upon our coding process and explores RQ2 by describing paradigmatic assumptions in relation to theoretical foundations, interpretations, and ontological relations. The framings are not mutually exclusive. Instead, they represent distinct yet complementary ways in which care has been approached and developed in HCI. As illustrated in Figure 7, the boundaries between framings are often fluid and overlapping.

[3]This can be visualised by filtering papers with positionality reflections in the timeline view on our website.

5.1 Care Framed as Management of Conditions

A prominent framing of care research in HCI is around managing or preventing health-related conditions. Research in this framing includes the most long-standing work on care in HCI, exploring the role of technology in health management in both clinical settings and everyday life. Publications here typically motivate their research through highlighting the prevalence of certain diseases and the challenges of care, while positioning technology as a potential solution. For example:

> *"Mental disorders are estimated to affect one in four people in their lifetime [...] yet the barriers to accessing mental health services today remain significant [...] Many have increasingly begun to view digital mental healthcare tools as a potential means of effectively addressing these challenges"* [42].

Definitions of Care. Although all papers used the term *care* extensively, studies addressing health conditions rarely defined it. It has been instead implicitly associated with the needs of specific health conditions (using terms like "cancer care" [118] or "patient care" [140]), or used to refer to a context (e.g., participation or partnerships "in care" [96]). What these exactly meant was often left for readers to interpret and can be highly context-specific.

The exception is a set of 12 studies on managing conditions at home, where self-care was explicitly defined. Earlier papers typically used the language of "patient" and defined self-care as an extension of clinical care (e.g., *"tasks a person has been requested to do outside a clinical setting by healthcare providers"* [1]). Later, this medicalised framing of self-care was criticised, and researchers started to emphasise its mundane nature, though still focused on chronic conditions. An important paper is Nunes and Fitzpatrick's ethnographic work on people living with Parkinson's, where they defined self-care as:

> *"the activities that people living with a chronic condition (patients and carers) undertake to manage the condition as part of their everyday life"* [148].

Corbin and Strauss's concept of the *"three lines of work"* [43, 44] in managing chronic conditions (i.e., illness work, everyday life work, biographical work) was frequently cited to explain what care involves (e.g., in [18, 19]). More recently, this definition has been further broadened, with less emphasis on specific conditions. For example, Capel et al. [32] described self-care as *"practices to maintain, recover, and improve one's health and wellbeing"*. A table of all the definitions is available in Appendix B.

Overall, there is an implicit negative sentiment around care in this framing, especially in earlier publications. Studies highlighted challenges associated with navigating healthcare systems and managing diseases at home. Being a hospital patient was described as *"one of the most dis-empowering situations one can experience in modern society"* [21], while hospital visits were considered *"stressful experiences for patients and their loved ones"* [159]. Similarly, care management at home was depicted as complex, challenging, and labour-intensive [1, 82, 96, 129, 136, 148]. One key challenge was to integrate medical care requirements and devices into people's everyday lives (i.e., non-clinical, multi-stakeholder, social spaces such as home and school) [1, 80, 151, 174]. While many studies

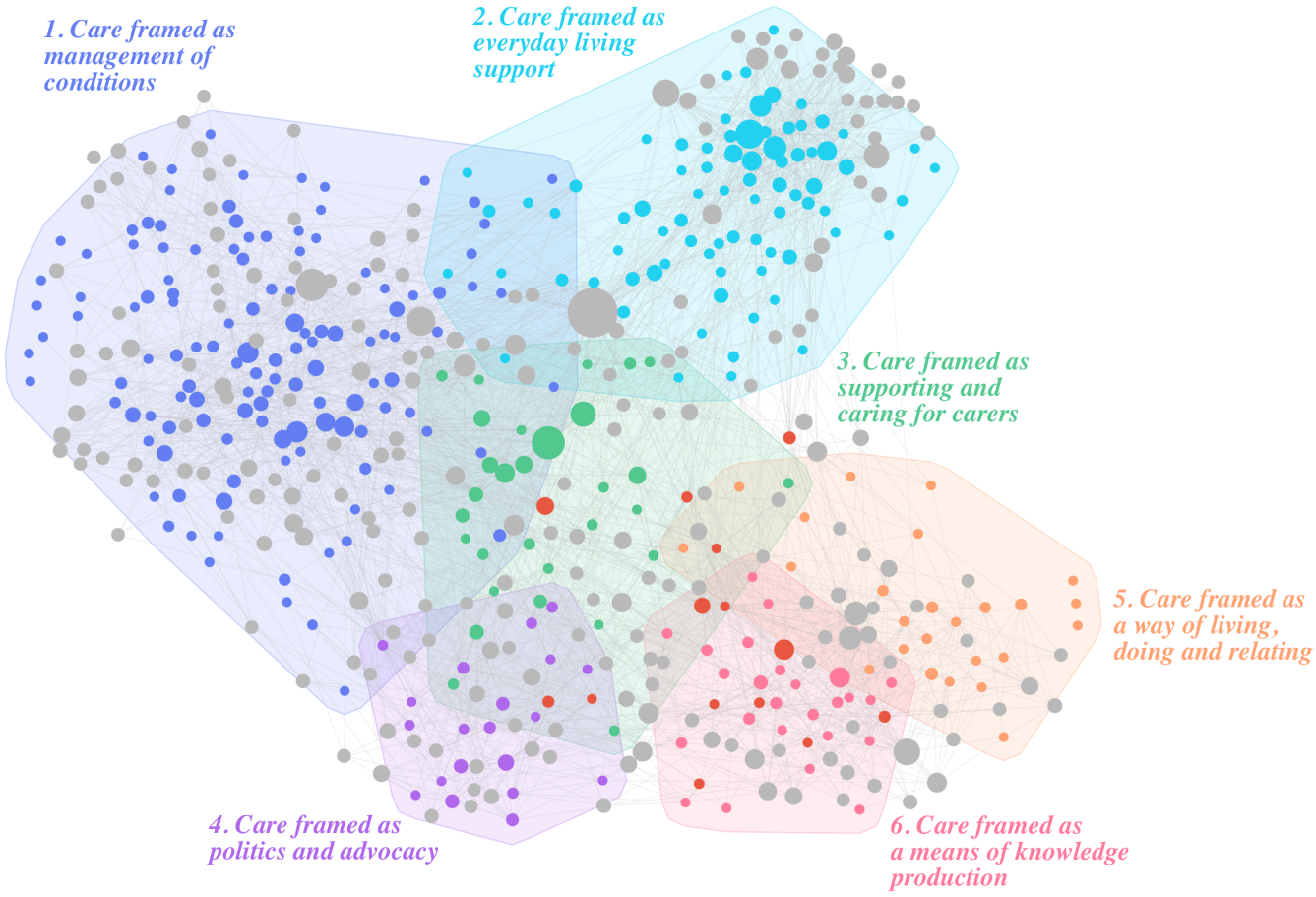


Figure 7: Visualisation of the reviewed papers, their key references, and the six framings of care. Details on the papers and references are available on our website: https://care-hci.org/.

emphasised the collaborative efforts needed for self-care—whether it is coordinating with family caregivers or communicating with professional health providers—they also placed considerable responsibility on individuals to manage their own health [96, 129]. This was evident in the description of hospital discharge as a process that *"transform[s] patients from completely passive recipients of care to being completely responsible for all aspects of their health care"* [21], and in portrayals of self-care as a strategy to alleviate pressure on healthcare systems by *"mov[ing] patients with incurable (or chronic) conditions from hospitals and clinics to the home, where they would self-care, or manage the condition by themselves"* [148].

Role of Technology. Technology design in this framing adopts a pragmatic orientation, emphasising improved health and care outcomes through human-centred design methods. Key values highlighted are agency, autonomy, and control over a health condition. In earlier publications, the role of technology was centred on information and communication related functions to help people more actively participate in their care and make better care decisions [118, 142, 159, 161]. Since around 2015, there has been a dual rise in research on the use of self-tracking technologies at home and on how patient-generated data could be used for clinical diagnosis [38, 42, 95, 102, 187, 197, 205], which might relate to the growing adoption of smartphones and self-tracking devices. As a result, patients' roles have shifted from information receivers from healthcare providers to those who generate and provide data. Alongside this is the broadening role of technology, from information sharing to value alignment, shared decision-making, and co-responsibility among clinicians, patients, and their care partners [18, 88, 104, 128, 129, 166]. More recently, there has been a rapid rise in research investigating how AI might help people make sense of complex medical data, assist with care decision-making, and support condition prediction and prevention [107–109, 121, 215].

5.2 Care Framed as Everyday Living Support

This framing centres care around the everyday needs of specific groups (with or without a condition). It is dominated by older adults and people with dementia (n=78), but also includes people with disabilities or in need of social care [62, 63, 163]. Papers are typically motivated by global demographic shifts, increased care needs, and care workforce shortages, while positioning the group needing care as a burden on a resource-limited care sector [27, 31, 34, 35, 64, 183,

196]. Ageing in particular is framed as a societal challenge, echoing discourses identified by Vines et al. [200]. Similar to the previous framing, technology is often proposed as a response to this care demand and supply paradox, and to enrich quality of life of those living at home or in residential care [33, 49, 78, 196].

Definitions of Care. Again, most studies did not explicitly define care. Often, care was implied as tied to meeting everyday needs and activities of the daily living of a population (e.g., supporting independent living or social communication) or the setting where it occurs (e.g., care organisations). This was also evident in three papers that did offer a definition, but all defined care via care work or the role of caregivers. For example, in their work exploring young carers of people with dementia, McNaney et al. defined care as:

> *"including a grandchild providing company and conversation to a grandparent, a youth who regularly looks after an older family member, or a youth engaging in formal care responsibilities within a care home"* [137].

There is a complex array of sentiments that come with this framing. Much of the literature emphasises the importance of care, and by association care technologies, in promoting independence, agency and quality of life. This is particularly evident in research supporting everyday activities for people with dementia. A lineage of this work builds on Kitwood's conception of personhood, emphasising dignity, identity, and relational care through biographical tools and artefacts [40, 71, 72, 201]. However, this framing is also dominated by perspectives that portray certain populations as a societal burden due to requiring additional care to support their everyday living. Studies have variously highlighted people's value of choice, control, and independence, while the perceived need for care was often associated with negative feelings [16, 22].

Role of Technology. These tensions around the sentiments are further reinforced when discussing technology. A common area of study has been the design and evaluation of technologies for ageing-in-place [22, 31, 50, 203] and assistive living [136]. A key tension lies between promoting *independence* and enabling *care* through these systems, as care to enable people to have more control over their lives can also increase feelings of disempowerment [79, 177]. Studies have also highlighted the impact of technology on family relationships and the new power dynamics it brings [52, 177]. For example, Vines et al. [199] observed remote monitoring systems reducing opportunities for meaningful family interactions and tacit care. Recent studies have begun to counter the dominant unidirectional models and narratives of care embedded in these technologies. This includes work that emphasises the interdependent, relational, and reciprocal nature of care in later life [165], and studies that involve older adults as active users and co-creators of technology to explore curiosity, play, and experimentation [184]. The connotation of care in these papers shifts from negative to more neutral or positive (e.g., *"care can inspire joy"* [165]).

5.3 Care Framed as Supporting and Caring for Carers

This framing centres on people who provide care through obligation or duty (jobs, roles, identities), including clinicians, nurses, community health workers, paid care workers and unpaid carers. It overlaps with the previous two framings in application domains (e.g., care for patients or older adults), but we identify it as a distinct framing because the focus shifts from improving the outcomes of care to considering the process, challenges, and impact of caregiving. This is clearly illustrated by Chen et al.:

> *"The goal of this paper is not to examine how caregivers can better devote themselves to patient care, but rather how caregivers can manage their own health and wellness in the process of helping others"* [36].

Definitions of Care. Most papers in this framing did not directly define care. Instead, their relations to care was established via a clarification on the varied roles involved in caregiving. For example, Long et al. [130] defined informal carers as *"those who provide care and support for another dependent person, often a family member, partner or spouse"*. Others defined the job of home health aides [141] and childcare workers [110]. Through these, we see that care is still largely framed as hierarchical (from caregiver to care recipient), rather than lateral (*"decentralized non-hierarchical networked systems of care"* [157].

Overall, there is a negative sentiment around the porous nature of care work. Though the work content is diverse, all papers describe care work as a complex practice, sitting across boundaries of private and public spheres, and involving a mix of physical and emotional labour [10, 36, 132]. For unpaid or family caregivers, this is reflected in challenges balancing caregiving roles with personal lives, work, and family. Studies highlight the value of peer support networks and *"temporary escape"* from caregiving [130, 210]. Yet, tensions have been identified between seeking help and the tendency to conceal it, which may be rooted in societal values of independence, moral obligations to care, and cultural norms of *"managing on one's own"* [170]. For formal care workers, porosity arises through blurred boundaries between paid and moral care, and between professional and personal roles. This porosity is closely tied to the much invisible yet essential labour that care work entails, a dimension increasingly recognised in recent research (e.g., [56, 141]).

Role of Technology. Although healthcare workers and care home staff have been involved as participants since the earliest studies on care, the central focus of technology has been on improving their work and care practices, rather than caring for them. Instead, these studies revealed how technologies typically disrupt their pre-existing care practices and generate more work [33, 78, 125, 160, 189, 214]. Recent studies turned their focus to care home workers as neglected actors in aged care, revealing the burdens associated with introducing and appropriating new technologies [34, 183, 204]. Similarly, for family caregivers, a recurring theme for technology is to support the above-mentioned access to peer support [130, 210], but rarely tackles the deeper societal factors that make sharing, connection, and escape challenging.

Starting in the 2020s, studies have turned to formal care workers outside clinical settings (e.g., frontline health workers and paid home carers), and started looking at the structural inequalities inherent in care roles (i.e, care as undervalued work that is disproportionately undertaken by women, people of minoritised identities and migrant populations). Technology is positioned with the potential to both reinforce and subvert these inequalities. For instance,

Tseng et al. [194] and Bartle et al. [10, 11] examined how technological interventions might improve equity for home care workers, whereas Lustig et al. [132] and Kaneko et al. [110] observed commercial childcare platforms reproducing unequal dynamics between clients and workers through asymmetrical interfaces, rankings, and surveillance features.

Two foundational references in this framing are Suchman's *Making Work Visible* [186] and Star and Strauss's work on *The Ecology of Visible and Invisible Work* [181], which underpinned much later work in critically interrogating the overlooked and underappreciated labour involved in care work. Over time, we see tensions around how technology should make care work visible. Early studies show that much care work happens behind the scenes and may stay invisible to co-workers and family members (e.g., medical secretaries and nurses' care for health records [26, 160], emotional labour and articulation work in caregiving [36]). These studies call for technology design to better account for such labour, and explore HCI's potential to *"make things public"* [130]. Yet recent studies note that visibility may not necessarily lead to the intended acknowledgement; instead, it can introduce additional surveillance and burden [141, 171]. For example, in cases where workers purposefully stay invisible to avoid state monitoring [157].

A further tension lies between portraying care labour as professional, clinical, and transactional—often to secure short-term material gains—and advocating for the broader, long-term legitimacy of care as work [141]. Designers must therefore attend to the consequences of visibility by asking what is made visible, for what purpose, and for whom [157, 171].

5.4 Care Framed as Politics and Advocacy

This framing retains the motivation around health conditions emphasised in the first framing, but differs in viewing the source of problems as structural and socio-political. While care may still be anchored in specific conditions or health needs, these papers incorporate feminist HCI [8, 9], social justice [47, 48, 60, 73], and post-colonial [100] theories to highlight marginalised topics, populations, and systems of inequality around care in society. Motivations include examining care subjects that remained under-explored or invisible due to social taboos around the condition or the group (e.g., [3, 4, 46]), and investigating how technologies might interact with sociocultural factors in co-shaping our understandings and practices of care (e.g., [155, 180]).

Definitions of Care. Three papers explicitly defined care (via intimate care [3, 176] and care infrastructures [6]). Others referred to care primarily in relation to health care (e.g., healthcare services, access to care, points of care), but did not clarify what they meant by that. Nevertheless, most papers in this framing extensively discussed what influenced and shaped people's understanding of care. For example, in their study of clinical practice around urinary incontinence, Almeida et al. emphasised that:

> *"In women, incontinence is 'neither an illness nor, strictly, a medical problem [...it is] a social rather than a biological pathology'"* [3].

Their work shows how intimate care practices are entangled with social taboos surrounding women's bodies and social expectations of *"a body in control"*. Similarly, Pendse et al. observed that experiences of mental health and the way people understand and articulate their feelings are deeply shaped by the societies they inhabit that:

> *"Social norms, laws, and our relation to institutions created to provide for or govern our well-being all have an influence on what kinds of care we believe is accessible to us, and how we go about finding care"* [156].

This is echoed by other studies showing that individuals are often reliant on, and subjected to, institutional assessments to validate their conditions and determine the care they can receive [3, 155]. However, this process may itself be imbued with political decisions. For instance, colonial power relations have historically influenced what forms of distress clinicians recognise as "mental illness", and what conditions have been ignored [155].

Likewise, research on maternal care in patriarchal and religious contexts emphasised the structural, social, and cultural influences on care practices. Studies revealed conflicts between Western medical knowledge and local culture, including religious beliefs, gendered power dynamics, and restrictions imposed by people's sociomaterial realities (e.g., living arrangements, mobile phone access, and health service availability) [6, 147, 188]. Many of these studies adopted ecological or infrastructural perspectives to unpack interrelated factors around care [46, 155, 176, 202]. As defined by Bagalkot et al.:

> *"care infrastructures do not only involve the 'structures, functions, and resources required' to provide [...] healthcare services but also all the physical, social and structural arrangements of dynamic human and non-human elements that support and sustain care practices"* [6].

Role of Technology. The role of technology in this framing is not to manage health conditions, but to improve the social conditions associated with their care. Papers highlight the potential of technology to facilitate access to care, improve health literacy, support bodily discovery and self-learning, and break down associated taboos [3, 4, 6, 188]. Key values include agency, equity, and knowledge, echoing earlier framings. However, this framing also foregrounds the tensions generated by social expectations of independence and control [3]. There is recognition that technology is not neutral and may reinforce problematic social constructions of care. For example, Spors et al. [180] demonstrated how commercial self-care apps reproduce individualistic and potentially harmful understandings of mental distress, while Pendse et al. [155] showed that colonial legacies continue to shape how digital tools define and respond to mental illness.

Studies also critique dominant interventions for individual behaviour change grounded in normative ideals of motherhood, showing that approaches focused on individuals inevitably collide with structural realities, such as overburdened health workers, opaque health systems, and gendered power dynamics within and beyond the household [112]. Drawing on notions of *"unsettling care"* [146]—which encourages looking beyond the "success" or "failure" of an intervention to examine how it configures relations in an already

gendered and unequal field—Karusala et al. [112] offered an example of thinking about these in practice, by examining a public-facing health literacy service in India.

Overall, this framing highlights the politics embedded in technology and underscores the need for a socio-ecological perspective when designing technologies for care.

5.5 Care Framed as a Way of Living, Doing, and Relating

This framing places care as a mundane, sometimes implicit, yet necessary organising force within all relations. Compared with previous framings, there is a move from human-centred to including more-than-human perspectives. The subject of care has broadened beyond people with conditions or in vulnerable positions to include virtual characters, living organisms, wildlife, forests, food, environment, and community. Along this is a connection to entanglement HCI [74], from seeing technology as a tool for care, to recognising technology as an actor in care that may itself need care. Though applied in diverse domains, a typical motivation of papers in this framing is the potential of technology to foster more caring and sustainable relationships. For example:

> *"sensing technologies provide a means to harness curiosity about the animals living in these spaces, and possibly also connect interest to care"* [198].

Definitions of Care. Papers in this framing more directly discussed what care means. While most papers centred on human relations did not cite specific care theory, they described it as *"mutual understanding and emotional support"* and *"visceral help"* when others are in need, which could be enacted through small acts like effortful communication and social touch [53, 106, 152, 168]. When studying human-robot interactions, Fallatah et al. similarly described care as:

> *"instances in which people go beyond help by making sure the robot exits correctly or wishing it well upon its departure"* [67].

Care is also framed as *"everyday labour of maintenance"* towards technological artefacts. Studies described various practical aspects of care (e.g., charging, cleaning, mending, and repairing the devices, or feeding the organisms embedded), but also highlighted the affective attachments and ethical obligations embedded in these mundane acts [12, 30, 69, 122]. Several studies [37, 77, 164, 213] drew on feminist care ethics, citing Tronto and Fisher's definition:

> *"care as a species of activity that includes everything we do to maintain, continue and repair 'our world' so that we can live in it as well as possible"* [192].

Others referred to STS scholar Maria Puig de la Bellacasa's concept of *"matters of care"* [133, 134], arguing that care involves a strong sense of attachment and commitment. These studies [41, 92, 213], many adopting a more-than-human perspective, highlighted care as an ethically and politically charged practice that *"we must take care of things in order to remain responsible for their becomings"*.

Overall, this framing conveys a positive sentiment towards care, emphasising its reciprocity and necessity. In contrast to earlier framings that depict care as challenging or burdensome, here, care is associated with joy, pleasure, connection, social intimacy, and commons. Studies have shown that people appreciate being able to both ask for and express care [53, 152]. Others highlighted the benefits of care-giving in fostering engagement with technology, family relationships, collective responsibility, and enhancing personal well-being [14, 30, 51, 87, 105, 117, 123, 154]. Nevertheless, many also acknowledged the need to balance appreciation of care and its potential labour or burden, stressing that care should remain optional rather than obligatory.

In this framing, care is also transformed from a context or practice to a value or virtue. Several papers highlight care as a key value that connects people, drives community activities, and underpins community-based governance [15, 55, 152]. These studies also position care as an alternative to capitalist modes of production, consumption, and exchange—offering a counterpoint to neoliberal priorities of efficiency and profit [7, 15, 55, 92]. They call for alternative economic futures that champion collective care and commoning, which Berns et al. [15] describe as the relational process through which communities negotiate access, care, and responsibility for shared resources.

Role of Technology. Unlike previous framings, the role of technology here is not to address specific care challenges, but to function as a medium that helps people notice and enact care. Although the specific functions differ, the common goal of these technologies is to break individualism and promote sharing, connection, and engagement. Studies on human relationships demonstrated how technology could offer an accessible channel and pre-negotiated mechanism for people to more easily express their care needs and demonstrate care to others [53, 106, 152]. In more-than-human relationships, studies showed that care-requiring could be an inspirational quality designed into artefacts. For example, Lu and Lopes embedded a living organism that requires care as a functional component of the device to foster human-tech relationships [131]. Others highlighted the potential of technology in helping people notice and attune to non-human beings. For example, Vella et al. used sensing cameras to encourage people to observe wildlife in their gardens and increase people's capacity to notice them, promoting acts of care [198].

Partly due to the mundane framing of care in this thread, many studies also argue that technologies should support caring experiences from ordinary, pre-existing activities and relationships, rather than creating something exceptional [20, 105, 115, 168]. In contrast to the emphasis on agency and control found in earlier framings, the values foregrounded here include curiosity, joy, pleasure, fun, vulnerability, interdependence, and connection [15, 20, 53, 106, 115]). Overall, we see a turn from utility and instrumental goals towards experiential gains and sustainability [20, 53, 115]. For example, Altarriba Bertran et al. draw on celebratory technology and ludic design, calling for re-orienting designers towards values of joy, care, and wonder [20]. Similarly, Berns et al. argue that design should be a *"collective endeavour enabling relationships and attachments to socio-technical resources"*, not simply producing artefacts [15].

5.6 Care Framed as a Means of Knowledge Production

This framing covers papers that use care as an analytical lens for understanding a subject of study, or as an ethical orientation guiding

the research process. Here, care is understood as a way of thinking and knowing, and central to knowledge production. It shares similarities with the prior framing in its relational conception of care, but differs in that care relations are not only a result of technology design (*"design for care"*), but also shape the design and research itself (*"design with care"*). It also extends the third framing on care work, from a practice to a lens that could be used to understand other practices and research sites. Because this framing is not confined to a particular domain or sector, its application areas and research motivations vary. Nonetheless, the papers share a common insight: reframing a topic through a care lens reveals aspects that were previously overlooked or difficult to address.

Definitions of Care. Many papers extensively discussed what they meant by care, often including a substantial related-work section on care as a concept (e.g., [2, 13, 113, 216]). Whereas papers in the third framing define care through the tasks or roles of caregivers, care work in this framing is defined more by its ethical orientation. For instance, it is described as:

> *"the continual affective and emotional adjustments that people make by noticing and attending to one another"* [13], which is *"characterized by an ethically-grounded orientation to human action centering on the relationships between the self and others"* [91].

Sharing a relational understanding of care, papers in this framing—like those in the prior framing, but to a greater extent—draw on Tronto and Fisher's and Puig de la Bellacasa's work on care. While the prior framing draws on them to highlight the necessity of caring relations, this framing recognises their inevitability, their uneven distribution, and the power dynamics that shape them. It recognises that technology design and research are themselves intertwined with, and constrained by, the unequally distributed care relations. As Akbaba et al. observed:

> *"matters of care are entangled with knowledge production, moving beyond 'responsible maintenance' towards a concern with exclusions, power dynamics, and relationships"* [2].

Therefore, care as a means of knowledge production is about attending to what, how, and when things get caring attention and come to matter, and what, how, and when they do not during HCI research [114, 133].

Compared with the largely positive tone towards care in the prior framing, researchers here foreground the politics and costs of care. Care is described as *"non-innocent"* [2], *"not unproblematic"* [120], and could produce *"discontent, unease, and agony"* [93]. Care is never neutral (i.e., to care for something always means not caring for something else), nor without cost (e.g., the emotional cost of attentiveness and conducting care research: *"the burden of constant and numerous points of reflection"* [113]). However, this framing suggests that awareness of these tensions could be generative for HCI research [2, 93, 150]. Drawing on Haraway's call to *"stay with the trouble"* [90] (i.e., remain with uncertainty and discomfort so that established beliefs and values may be challenged) and Murphy's notion of *"unsettling care"* [146] (i.e., the purposeful undoing and troubling of particular arrangements so that they might be acknowledged and remade in less violent, more liveable ways), these papers resist romanticising care or accepting simple quick solutions. Akbaba et al. [2] argued that grounding care in the complexity of politics, power, and interpersonal relationships constitutes a form of situated knowledge-making, reflecting the value of what Haraway called *"partial perspective"* [61]. This turn in epistemology, together with greater reflexivity about research and design practice, has led to a growing number of studies adopting first-person or autobiographical methods.

Role of Technology. The role of technology is very different in this framing from others; it is less about technology to support care, but using care as a lens to study technology. Papers do this in various ways. One thread of papers apply care through the lens of "care work". They share the political associations of the *caring for carers* framing—highlighting its gendered division, invisible and undervalued history. Domains such as everyday philanthropy [91], gig work [59], and computer security support [195] are positioned as care, revealing invisible or taken-for-granted qualities of the underlying work. As Harmon et al. highlighted:

> *"a reframing of philanthropic work as care work helps to open up the design space for philanthropic technologies [...] from streamlining resource redistribution and towards supporting more diverse social practices"* [91].

Another thread is where authors position design or research itself as care work and as matters of care to both researchers and participants. For example, Spiel et al. [178] and Michelson et al. [139] revealed the care work and personal values embedded in making design artefacts. Others position care as a way to navigate dynamic relationships between researcher and participants [98], their design materials (e.g, [99, 114]), and as a way to cultivate ethical sensitivity during design practice [76, 173]. Together, these scholars highlight care as an under-acknowledged element of social relations that is crucial to understanding the broader ecosystems in which technologies operate.

This dialogue is both enriched and complicated when applied to collaborative projects involving multiple stakeholders who all care for an issue or group, but may care differently [2, 86, 120, 167]. This complexity foregrounds questions regarding what we care about, and how we care, in design and research. A common thread across these studies is their attention to different temporalities. Krüger et al. [120] and Akbaba et al. [2] examined the maintenance of design artefacts after project completion, while Sabie and Parikh [167] highlighted the importance to make *"care time"* in cultivating relationships among students and the community they designed for. However, time is not without tension. Halperin and McElroy [86] described the need for both *"slow design with care"* and *"rapid design in response to the urgency of crises"*, and the trade-offs they made between their values and the project reality.

This is also reflected in other studies where researchers' willingness to "care" is often shaped or constrained by structural factors in academia and society. For instance, Krüger et al. [120] showed how caring for projects often meant additional work beyond funding periods and can be at odds with normative expectations of novelty in HCI research (e.g., maintaining a website needed by partners). Akbaba et al. [2] further revealed how this currently devalued care labour in research typically falls to students or early-career researchers, drawing attention to power asymmetries. Taken

together, we can see a need for more caring practices in design research, whilst acknowledging the socio-political realities under which the research is conducted.

6 Transformative Potentials of Connecting and Mapping the Field

Our review demonstrates the breadth and diversity of HCI research on care. As seen in the six paradigmatic framings, care takes on many definitions, has a wide scope in terms of intended impact, and an array of interests in and implications for technology design. To further synthesise our findings, we mapped these framings into Figure 8 and 9. The making of the map enables articulation of the connections and tensions across framings, and supports discussing the temporal development of the field in relation to HCI. As highlighted by Sanders [169]: *"Making a map is a way to hold a domain still for long enough to be able to see the relationships between the various approaches, methods, and tools"*, and it is useful in demonstrating the complexity and change of a field.

Overall, we observe a broadening and deepening of the role of care in HCI, **from viewing care as a problem and technology as the solution** (i.e., the negative connotation of care, as evident in our first four framings), **to recognising care as a generative and emancipatory lens for technology** (as shown in the last two framings). Figure 8 represents this by positioning the framings across horizontal and vertical axis that attend to the underlying conceptualisations of care that thread across the framings (horizontal) and what is intended to be targeted, understood and changed by researchers (vertical). The map reflects the understanding over time: from care being driven by the needs of health conditions (left, horizontal axis) to being seen as necessary in sustaining all relations (right, horizontal axis).

It also reveals the evolving scope of HCI research, from understanding and supporting in practical terms individual or collaborative capacities to care (top, vertical axis), to examining how broader social, cultural, political contexts shape care practices (bottom, vertical axis). Along this trajectory is the expanding focus from individual actors and their microsystem (e.g., patients, clinicians, caregivers) to consider wider care networks, infrastructures, and ecosystems. This also comes with a growing recognition that technology is not neutral but inherently political, entangling with and co-shaping our socio-political realities and care experiences. This includes a turn inwards to consider how HCI research itself is embedded within structures that shape caring for research, researchers, and participants. Placing a layer over this map representing the development process, one could read the map as moving from the upper left towards the bottom right corner over time (see Figure 9).

6.1 Mapping through Care-Tech Relations

The evolving understanding in the literature is also revealed in the role of technology and its relationship with care. As a means to further draw connections and differences between framings, and their developments over time, we have layered onto the map four types of care-tech relations identified across our framings (shown in the quadrants in Figure 9). Once again, illustrated by the dotted

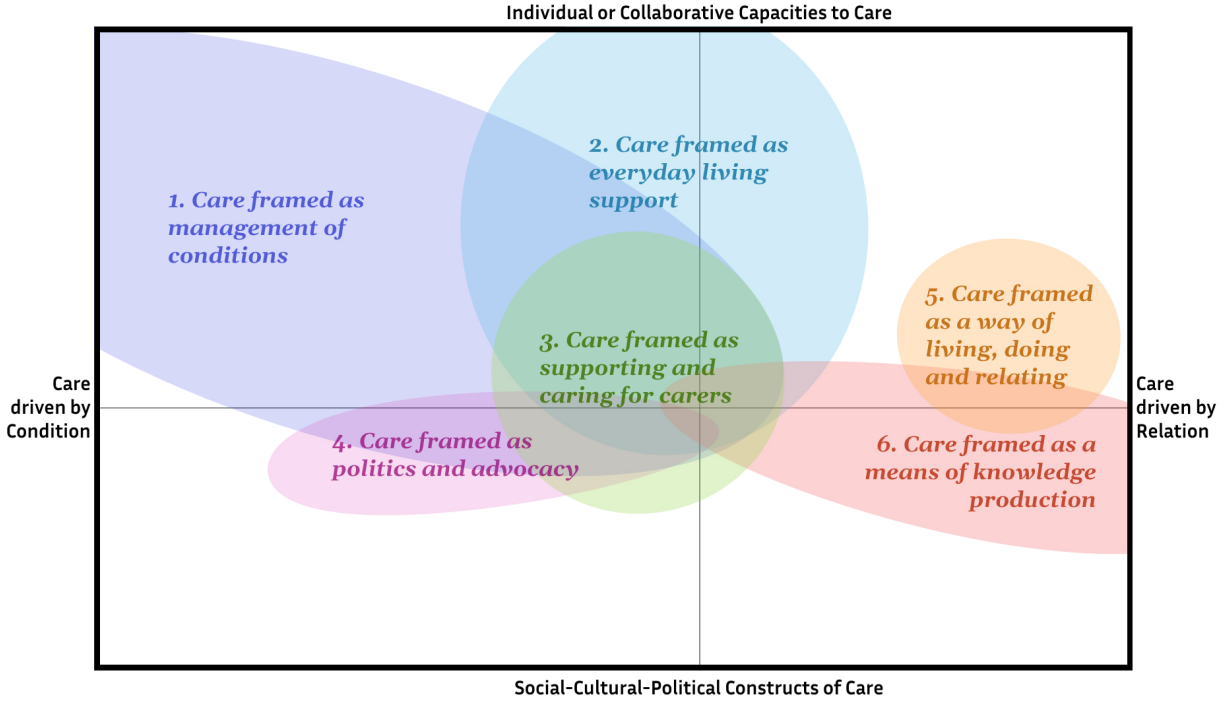


Figure 8: A conceptual map that connects and positions the six framings of care. Note. The map is structured along two axes: from condition-driven to relation-driven care, and from individual or collaborative capacities to socio-cultural-political constructs of care.

lines, we see the broadening role of technology from being a means to improve care, to redefining conditions of care, to itself needing care and being embedded in care relations, and finally towards care as a way to redefine technology design and research. We explain the quadrants further below.

6.1.1 ***Tech as a Means to Improve Care.*** The first quadrant on the map marks the early traditions of HCI research on care, where technology is seen as a means to improve care or address challenges posed by health conditions and ageing. This quadrant shows the commonalities among the first three framings of care detailed in Section 5. Here, care is primarily portrayed as practice, work, and labour—often with negative connotations—that technology could support or reduce. The value of technology lies in empowering individual or collaborative capacities to care (e.g., through information, knowledge, and skill). Though targeted at various settings and actors, the three framings within this quadrant reveal a consistent emphasis on the collaborative and ongoing nature of care, the need for technologies to fit into existing practices, and consider the social and material contexts within which care happens.

However, the scope of the context differs from the lower quadrants on the map, in that most studies did not engage with the broader sociocultural or political context that practices of care exist within, nor do they make explicit the researchers' positionality. While many findings point to social, cultural, and institutional factors, the conclusions of studies in this quadrant tend to be limited to improvements of existing designs, or better personalisation and adaptation of a technology (e.g., [27, 196]. Underlying these may be the assumption that solutions can be reached if technology is given to or oriented towards the "right" people (e.g., for clinical care to be better, staff need better tools; formal care is overburdened, so equip individuals to manage themselves).

6.1.2 ***Tech as a Way to Redefine Conditions of Care.*** Moving below to the second quadrant on the map, this includes research that framed care as politics and advocacy and as supporting and caring for carers (especially the later half of this framing, as explained in Section 5). While care is still driven by health and care needs, work in this quadrant recognises that conditions do not exist in isolation; they are constructed by social, cultural, and political embeddings. Therefore, the value of technology here is to redefine the social conditions of care needs or caregiving. This includes understanding technology as value-laden, which could both reinforce or subvert social constructs. In this way, it extends the first quadrant in emphasising that technology may also be part of the problem in maintaining definitions of a condition. Indeed, the condition may be the wrong foci for a technology to address, such as in normative sites of motherhood. This offers alternative pathways for designing technology to address care needs driven by health conditions, and invokes a necessary political and critical lens when designing technology for care.

6.1.3 ***Tech as Part of Care Relations.*** The third quadrant is represented by research that frames care as a way of living, doing, and

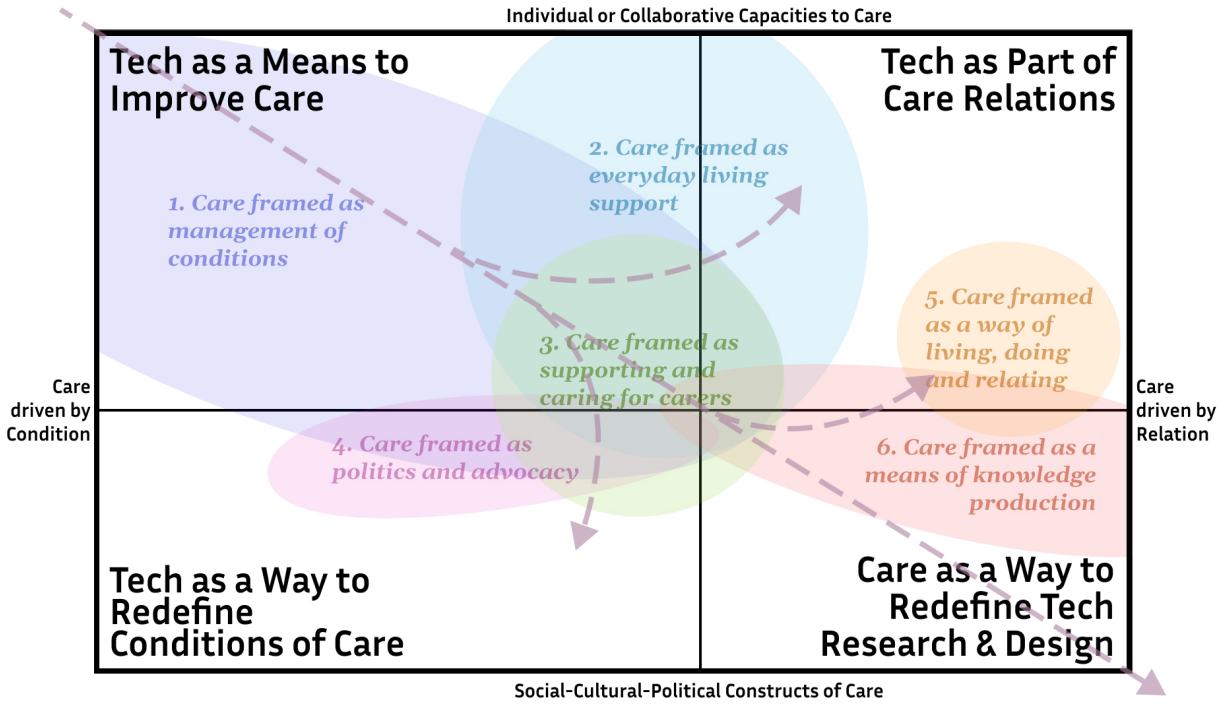


Figure 9: The conceptual map of the six framings of care, highlighting the temporal development (from the upper left towards the bottom right corner) and the quadrants representing four types of care-tech relations: 1) Tech as a Means to Improve Care, 2) Tech as a Way to Redefine Conditions of Care, 3) Tech as a Part of Care Relations, and 4) Care as a Way to Redefine Tech Research & Design.

relating, and the turn to relationality in care to support everyday living and caring for carers. In this quadrant, technology is not only a tool applied *in* care, but itself becomes entangled and part of the caring relations. Notably, while studies of the framings in this quadrant began to emerge around the same time (mid-2010s), their origins came from very different places. The work from the second and third framings emerged as a response to earlier, more technologically driven efforts to design technologies for remote care, or to augment and replace certain forms of care work, recognising that technologies should mediate and support social relations rather than replace them. However, technology here is still viewed pragmatically as supporting relational care. Instead, and in parallel, the fifth framing was mostly inspired by wider theoretical developments outside HCI, and highlights an alternative view of human-tech relationships (i.e., from using to caring), offering examples of how care could be a quality designed into technology itself. However, while framings in this quadrant adopted a broader understanding and recognised networks and ecologies of care, there was relatively little attention to the political and cultural contexts where the study was conducted and associated systemic inequalities. This differentiates it from the lower quadrants.

6.1.4 ***Care as a Way to Redefine Tech Research & Design.*** The fourth quadrant turned to care as a way to redefine technological research and design, and is represented by the final framing. Similar to the previous quadrant, a relational focus on care is embraced. It differs in that care is used as a theoretical lens to critique, understand and re-imagine technologies, which is central to knowledge production. The subject of study also turns from a focus on designing and researching technology for others towards the research and design process itself. Here, care is no longer designed *for*, but designed *with*. Moving from the third to the fourth quadrant involves a resistance to the romanticisation of care: shifting from seeing it as inherently positive to the awareness that care is not benign, never neutral, and always comes at a cost (e.g., attention, labour, resource). This also shifts the goal of HCI research from designing technology to solve problems in care (the left two quadrants) and imagining new interactions that foster caring relationships (the upper right quadrant), to unsettling, revealing and staying with the trouble [146].

6.1.5 ***Implications from the Mapping.*** In bringing together diverse literature under six framings of care and further positioning them onto a map, it is possible to see both synergies and overlaps, as well as inherent tensions and contradictions, across care literature in HCI. Based on these, we draw out four implications below.

A first implication is that what is meant by a condition for/of care is not uniform. Tensions in care-tech relations across the left two quadrants highlight the differing scale and scope of the meaning of a condition: e.g., physiological problem or state-of-wellbeing to the structures that define how these conditions came into being or are experienced through technology. To navigate this tension, the value of the framings in the left two quadrants is to clarify the role of technology and what is aimed to be changed by research (e.g., making the human experience better in hospital, or challenging a social taboo). Defining care here is about specifying context and politics, i.e. what, at a practical level, technology might change to make a condition "better".

A second implication for future research is that care as an organising force is also not unified. Tensions in care-tech relations across the right two quadrants highlight differences between care as an everyday situation that maintains relations and a generative lens to redefine relations. To navigate this tension, the value of the framings in the right two quadrants is to clarify the role of care and how it is conceptually situated (e.g., as a positive quality to design into technology, or as a critical lens for understanding technology). The role of care in these quadrants extends beyond a particular functionality of technology to include, for example, research itself.

A third implication is the differences between the types and temporalities of the impacts from care. Tensions across all quadrants highlight clearly defined and measurable impacts of technology in a particular application area against the unmeasurable unsettling of structures and raising issues elsewhere. While the former does work in attending to the urgency of specific situations, the latter aims for longer-term change to avoid such situations in the future. Yet there is a risk that the latter (right quadrants) becomes too conceptual or theoretical when it turns inward, losing the specificity of the external situation or an application area.

A fourth implication is how to resist resolving or removing the above tensions, and instead approach them as generative resources. This includes engaging with how the different framings could learn from each other and better position themselves. For example, the framings highlight different assumptions between *care as exceptional or abnormal* and *care as mundane and necessary*. While the first three framings addressed different problems (e.g., paradox between the care demands and care supplies, or people's desire for independence and need for care), they similarly assumed that care is needed when things deviate from normal, and that care restores the normative ideals of a healthy, able, independent person. The fourth framing challenged this by recognising care as problematic because of the social constructs around it, rather than it being isolated issues to be resolved or individual behaviour to be changed. While the fifth framing reveals the positive potential of care, the final framing reminds us that this positivity is not without cost, nor distributed evenly. We see these tensions as indicating potential pathways for reframing the intent of care and its scope of impact. For example, this might be through redefining a problem or ideal related to care from being a matter to be resolved between people to a matter of broader societal or political concern.

To offer an additional example, what is presented as a burden or challenge to carers' well-being in the third framing is reframed in the sixth as generative and desirable: a way of understanding work that extends beyond the transactional and draws attention to previously overlooked yet important aspects of sociotechnical systems. The implication of the latter framing on the former is that the porosity of care work is not necessarily something to avoid, but can be recognised or embraced in the design of technology. The implication of the former framing on the latter is a reminder of the potential unevenly distributed labour and politics associated with care work. We see the generative potential of this tension in how the perceived presence or emergence of a boundary can reveal, for example, overlooked settings or labour, and provide opportunities for navigating boundaries through porosity.

In summary, tensions across framings highlight the necessity to not omit what was previously included or defined as care, but shift

our practice by reframing what counts as a context, problem, site of care, or technology research.

6.2 Connecting with Four-Waves of HCI

Our findings on care also aligns with wider developments across the three waves of HCI [23, 24] and the more recent discussion of a fourth wave [5, 74]: *moving from a focus on workplace usability and efficiency* (i.e., in the clinic), *to everyday user experience* (i.e., care technologies at home, or between clinic and community), *to the ethics, values, and politics of technology* (i.e., care as a critical lens). Moreover, we see that HCI research on care not only exemplifies this movement through waves, but has the potential to advance it, offering practical insights into what fourth-wave HCI might entail. Indeed, many challenges highlighted in the first four framings in our review can be seen as tensions where *"third-wave HCI [methods] meet fourth-wave challenges"*. These challenges partly align with the findings from previous reviews on specific application domains of care (e.g., organisational and collaborative barriers related to clinical work [70], social and contextual factors in managing chronic conditions at home [149]).

With a fourth wave in mind, our review reveals the value of care in HCI research as a way to:

(1) alter human-technology relationships towards a more caring one;
(2) foster our sensitivity towards the marginalised and easily neglected actors and aspects—which may otherwise only be a secondary focus in research—by asking who is affected by and who benefits from technology;
(3) attune us to the wider institutional and structural contexts that constructs a condition or relation, and where individuals and technologies are embedded within and affected by.

Along the orientation, there is a move from objective to situated knowledge [89], and the application and development of first-person approaches (e.g., [93, 113]) and greater reflexivity and transparency of positionality during research [76]. These orientations (both theoretical and epistemological) point towards fourth-wave HCI, seeking not only to transform practices through technology or individual-level interventions alone, but to reshape social realities and infrastructures through relational and structural change.

7 Moving Forward, with Care

As demonstrated in our analysis, care is not uniformly understood, employed, or contributed to within HCI. Thus, our review can be seen to support care as intrinsic to the everyday fabric of life, including technology research and design, while simultaneously slippery, elusive, and multivocal in practice. This tension between care being necessary, yet divergently present in intent and impact, grounds care as transformative [134]: *its presence in one form, as demonstrated by the six framings, foregrounds its absence, difference, or possibility in another form.* This is exemplified through our mapping and discussion in Section 6.

The explicit presence of care through the articulation of definitions from specific studies and within identified frames is useful for researchers to position their work in the community, including situating its relevance, intent, and impact. Yet the transformative potential lies not only within definitions of care—i.e., their clarity and alignment in a knowledge-building community like HCI—but also in drawing attention to what may be missed or neglected when aligning with one frame or another. As such, the framings, along with the definitions, sentiments, and care-tech relations, provide possibilities for both a positioning of research or articulating of an audience, and a challenging of the framings themselves through new definitions of or through care. As highlighted, the aim of this review is not to erase tensions, but to foreground existing definitions and possibilities of their ongoing evolution.

The value of the paper and the accompanying map lies in their ability to momentarily hold the field in place—offering a pragmatic snapshot for understanding. As noted, while the map reflects the field's current proportions and positions, it should not be seen as fixed, but as a tool to think with, helping the field evolve and trace its development over time. The benefit of the visualisation tool lies in its ability to show the intricate relationships between papers over time and beyond the linear constraints of a written text. It is both a navigation tool and methodological resource which opens up ways of interpretation for other readers, and an epistemic device and an ongoing site and concrete mechanism through which the paper's *"transformative potential"* can continue to unfold. We encourage researchers using the tool or our dataset to further explore the field.

7.1 Complementing and Building on Previous Reviews

Our review responds to calls for policy-level, cross-disciplinary perspectives on care [70, 149], and extends previous reviews by analysing care across domains and over time. In particular, our analysis and mapping complement Petterson et al's recent integrative review [158] of a subset of this literature in two ways.

First, while our framings of care share similarities with the six conceptualisations identified by them (discussed below), we extend their work by organising these framings along two axes and situating them temporally and spatially, linking to their relationships with technology and to waves of HCI research (Section 6). Through this, we provide practical guidance for future HCI work. Secondly, whereas Petterson et al. propose an abstract tool (the flower diagram) to encourage care ethos in care practices, our work details the types of practices and ethos adopted within the community to date. We further extend their work by systematically analysing who, what, where were involved or excluded in care research, their study methods and traditions (Section 4), and offering concrete examples of how different framings and practices might inform and strengthen one another (Section 6). Our online tool offers both a timeline view and a graph view, presenting each paper with its metadata, framing of care, and co-citation links. This further supports researchers in interpreting relationships across the literature and in identifying connections between fields and research agendas that might otherwise appear unrelated.

The framings of care we identified both overlap with and differ from those in Petterson et al.'s review [158]. For instance, their conceptualisation of *care as a service*—"structured and professionalized set of activities aimed at supporting the wellbeing, health, and quality of life for individuals"—appears across our framings of care as the management of health conditions, support for everyday living, and caring for carers. We separated these areas to draw out the

specific characteristics of each research tradition. This also incorporates self-care and informal care, which were largely absent from their conceptualisations due to their focus on "networked forms of care" in the search terms. Similarly, their conceptualisation of *care as a burden* was found in our coding of sentiment. However, we did not group papers according to sentiments, but intend to illustrate how attitudes towards care shift over time. Our analysis revealed the deepening in understanding of care from *negative connotations* (e.g., care as a burden or challenging), to *positive* (e.g., care can inspire joy), to *political* (e.g., care is not always benign and not un-problematic).

We highlight these commonalities and differences not just to emphasise the contribution of this paper beyond Petterson et al's, nor to suggest that our approaches are incompatible, but to argue that there is a generative opportunity to have both views. This aligns with our intention for a pluriversal understanding of the field. Seeing these together moves care research in HCI towards a more comprehensive and synergetic understanding of not just how the field drew on conceptions of care from other disciplines, but also helping us work towards a theoretical framework of care that is underpinned by HCI research. Furthermore, this embraces our feminist sensitivities to how all knowledge, and therefore analyses like ours and Peterson et al's, are partial and formed from a position [61]. Discussing them together acknowledges these different positions while helping to bring this diverse literature together.

7.2 Research Gaps & Future Directions

Our review also reveals a number of research gaps in the field. We discuss the gaps and suggestions for future research below.

7.2.1 ***Bridge Care Ethics with Core Care Contexts and Look at Neglected Actors in Care Research.*** Despite its breadths and diversity, our analysis shows that who and what are given caring attention in HCI is unevenly distributed. We found that studies largely focused on older adults and people with chronic conditions, and were predominantly situated in formal care settings. While this reflects long-standing traditions in care research, these populations and settings were featured far less in the latter two framings of care. Echoing Waycott et al. [204], our review suggests that care ethics is most often used as a lens in contexts where care is covert or implicit, rather than in settings where care is a primary activity. This indicates a gap between care domains and care theories: technologies designed for care contexts are rarely considered through a care lens, and care ethics tend to be applied at a distance from domains where care practices are central or critical. Such gap risks diminishing the significance and impact of care in HCI. We suggest that future researchers more carefully attend to who or what receives caring attention within each framing, and to work towards bridging these framings of care. One promising direction is to use *care as a means of knowledge production* to make visible those groups that remain peripheral or overlooked.

7.2.2 ***Diversify Geographical and Cultural Perspectives on Care.*** This uneven distribution is also evident across study regions. Our analysis showed that care-related research was largely dominated by Anglosphere countries. While care ethics calls attention to marginalised and neglected perspectives, most studies in the review (especially the last two framings) were situated in well-resourced contexts or Western settings. Only a few papers explored other cultural contexts, and mostly limited to healthcare access, maternal care, and aged care. One factor may be linguistic: the concept of "care" does not always translate directly. For instance, in Chinese and Turkish[4], care as a concern and as a practice are two different terms, and the English notion of "care" as a multi-layered concept does not apply. This partly explains why studies in non-English speaking regions rarely adopt "care ethics". This is also a limitation of this review, that by focusing on papers explicitly use the term "care" or "caring", and by including only English-language publications, other relevant perspectives may be excluded. Murphy warned us how matters of care could intersect with postcolonial, transnational, and racialised contexts [146]. Our review foregrounds this disparity, and the political conditions within which care already circulates. As care is necessarily cultural and context specific, we call for more research on other cultural contexts, and adapting or broadening our understanding by incorporating other theories and cultural knowledge.

7.2.3 ***More Theoretical and Methodological Development.*** Our analysis in Section 4 revealed a lack of theoretical and methodological contributions, echoing the distribution of CHI publications in 2016 [208]. Our analysis of care framings further indicated that while researchers claimed to adopt a care ethics or a care lens, there was little agreement on what this means for practice, and at times, the approaches and methods adopted appeared closer to a conventional user study. We see the value of methodological clarity and theoretical development in two ways. First, it can encourage more deliberate adoption of care as a lens; secondly, it can prevent care from becoming a buzzword or merely a moral stance, and ensure it translates into practices to retain its transformative potential. Our review offers an initial step towards this. Section 4 illustrated alternative research approaches (e.g., situated knowledge, first-person perspectives, and a broader range of analytical techniques), and how care might support reflexivity during the research process. Meanwhile, discussions of the framings in Section5 showed how care as a quality and lens might be useful for HCI design and research. Our mapping (Figure 9) and website (https://care-hci.org/) further visualised and instantiated the paper's argument by making visible the relationships between different framings, care–technology relations, and the temporal development of HCI as a field. Through this, we both highlight how care-centred research might be conducted differently, and underscore the need for further methodological and theoretical development to support *"designing for and with"* relational understandings of care.

8 Conclusion

This paper presented a meta-narrative review of SIGCHI publications on care. The analysis of actors, contexts, study regions, technologies, and methods involved illustrated the landscape of care research in HCI, drawing attention to what may have been excluded or overlooked. The six paradigmatic framings of care synthesised from our analysis provided a foundational vocabulary for dialogues across research traditions. The conceptual map and

[4]The aspect of Chinese is based on the first author's personal experience, and the aspect of Turkish is shared by a visitor to my previous public exhibition on care.

the accompanying website further connected and positioned these framings in relation to technology, serving as a tool helping researchers navigate, position, and articulate care in their work. We concluded by calling for more cross-domain applications, theoretical expansion, and methodological development in future research. Together, we envision this work as a theoretical and methodological resource for researchers across diverse care domains and research traditions to build upon in situating, clarifying, or expanding care.

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Author Contribution & Background. (Melody) Zixuan Wang led and conducted the literature review and writing of this paper. She also initiated the visualisation exploration of the literature and designed the website. She is a PhD candidate within the Advanced Care Research Centre and the Institute for Design Informatics. Her PhD explores methodological guidance for designing with and for care in everyday later life.

Yuanrong Guo was involved in the one-day pilot coding workshop, first-round coding and synthesis of the clinical care and healthcare related papers. She has also contributed to some other coding in section 4 and made Figure 2, 5 and 6. She is a PhD candidate in Design at the Edinburgh College of Art, University of Edinburgh. Her research explores how designers engage with systems thinking in complex healthcare contexts.

Eilidh Bowman was involved in the paper screening, one-day pilot coding workshop, and first-round coding of the more-than-human thread of papers. She is a PhD candidate within the Advanced Care Research Centre and Design Informatics. Her PhD critically examines companion technologies for ageing-in-place.

Yuxiang Zhai was involved in the paper screening, one-day pilot coding workshop, and first-round coding of the meta-categorisations (Section 4) of the aged care and dementia care thread of papers, which aligns with her research expertise regarding on design for ageing population. She is a PhD candidate at Tsinghua University, and her research focuses on understanding how speculative design can be applied for design practice with older adults.

Xinhuan Shu did the front-end development of the care-hci.org visualisation website. She is a lecturer of data visualisation at Newcastle university. Her research focuses on designing human–AI collaborative systems to democratize how people engage with data through visualisation.

Shengchen Zhang did bibliographic data processing for the website and built visualisation prototypes for early exploration of the data. She also improved the visual design and accessibility for some of the figures. She is a PhD candidate at Tongji University. Her PhD explores dialogue design that builds mutual conceptual understanding between humans and robots. She also maintains several open-source tools for literature search and organisation.

Karey Helms was involved in the review synthesis and paper discussion. She drafted the related work section, helped to refine and steer the research questions, and contributed to the discussion of the implications and transformative potentials of the review. Many of her insights weaved nicely into the paper. She is an associate professor at Umeå Institute of Design in Sweden. Her research draws upon care ethics and posthuman feminism to design for situates where technology is often considered out of place, such as in outdoor environments and intimate settings of care.

Tara Capel was involved in the self-care and care definition synthesis workshops. She joined Melody's PhD supervision team from August and gave valuable suggestions on the writing of the method and findings (Section 5). She is a lecturer at the University of Edinburgh in the School of Informatics and Institute for Design Informatics, and her research draws upon participatory and feminist approaches to study and design technologies that support health and wellbeing.

John Vines was the initiator and supervisor of this project. He was involved in the one-day pilot coding workshop and synthesis of the framings, and helped double-review one-third of the papers. He contributed substantially to the initial formation of the six framings, as well as the reviewing and revising of the manuscript. He is a Professor at the University of Edinburgh in the School of Informatics and Institute for Design Informatics, and works across care, health, finance and civic technologies using participatory methods.

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A Coding Schemes

Table 2: Coding scheme.

Group	Code	Definition
Framing of Care	Study Motivations	How the paper framed the motivation of the study, typically at the beginning of the abstract or introduction.
	Definitions of Care	Whether the paper gave any explicit definitions around care and via which concept it was defined.
	Descriptions of Care	Since most papers did not define care, we also coded any descriptions directly on care to help decide its care framings. This includes 1) practices or attitudes included in care (e.g., taking the medicine, feeding the organism, emotional attachment); 2) subject of care (e.g., care for who or what); 3) qualities of care (e.g., care requires knowledge and skill, care is a collaborative effort).
	Key Concepts or References	Key concepts or references used when discussing care.
	Sentiment around Care	We further coded any sentiment around care indicated by the paper. These were grouped into four categories: 1) care as generally negative (e.g., described as challenging, complex, burdensome, or demanding); 2) care as generally positive (e.g., described as necessary, important, or associated with joy and pleasure); 3) Care as generally neutral (where no explicit sentiment or descriptive terms were attached to care); 4) care as political (e.g. statements such as "care is not always benign").
Care-Tech Relations	Role of Technology	The role of technology in relation to care (e.g., whether it is designed to support a specific care needs, foster caring relations, or itself needs care).
	Values Highlighted	The values highlighted in the technology design.
Tensions, Gaps, Implications		Any tensions, gaps, or implications identified in the paper.

B Definitions of Care

Table 3: Definitions of Care

Ref	Year	**Quote** (References in the original text have been omitted to avoid confusion with those of this paper.)	**Define via**
[1]	2010	We use the term self-care to denote the tasks a person has been requested to do outside a clinical setting by healthcare providers. [...] While we use the term 'self-care' to contrast the work done in the home to that of the hospital, we do not consider self-care to be an individual task and neither to be tied to the home.	Self-care
[83]	2013	Mallery et al. define prevention (in a healthcare context) as any action that "delays disease or slows its progression".	Preventive care
[17]	2017	Corbin and Strauss named three "lines of work" to [...] We use the term self-management to refer generally to these lines of work, but we acknowledge there is inconsistency regarding the definition of self-management and related terms (e.g., selfcare, [...]). Wherever possible, we ground our discussion in the activities themselves instead of referring to self-management or self-care.	Self-care
[148]	2018	Self-care is defined here as the activities that people living with a chronic condition (patients and carers) undertake to manage the condition as part of their everyday life. Engaging in self-care may occur in complement to the medical care undertaken by healthcare professionals, but is necessarily different from it.	Self-care
[129]	2019	Often with help from family caregivers, these individuals perform the daily work of self-care, which can include: illness work (e.g., physical activity, taking medications, monitoring blood glucose levels and blood pressure), everyday life work (e.g. obligations to family, job), and biographical work (e.g., coping with the emotions around accepting and adjusting to changes in life due to illness).	Self-care
[64]	2019	In defining self-management, Barlow's definition is used, which describes self-care as the ability of the individual to manage symptoms, treatment, emotions and lifestyle changes as part of living with a chronic condition.	Self-care
[58]	2021	Researchers in the HCI literature define self-management as "manag[ing] the symptoms, treatment, physical and psychosocial consequences and lifestyle changesinherent to living with a chronic condition" [...].	Self-management
[180]	2021	Self-Care can be described as an individual's desire to look after themselves, by themselves, recognisant of "[...] a more holistic conceptualisation of the self that relies on a connection between the mind and body". A practise ofself-care is therefore aimed at the maintenance of personal "wellness" and being "well-functioning"; with motivations for it ranging from "[...] addressing medical or mental health needs to simply relieving daily stress".	Self-care
[180]	2021	In itself, self-care can be understood as a routine activity to holistically manage, maintain and make sense of one's (mental) health: It is a very personal, but "mundane" endeavour, sitting at the intersection of medical, personal and private spaces; somewhere between preventive health measures, personal improvement and introspection, individual actualisation and practised spirituality.	Self-care
[39]	2022	Self-care is defined as the activities that someone takes "to lead a healthy lifestyle; to meet their social, emotional and psychological needs; to care for their LTC; and to prevent further illness or accidents" [...]. Self-care may incorporate aspects of self-managing medical treatment provided by HCPs for clinical care – 'to care for their LTC', but, after it is defined herein as broader than LTC self-management to involve a person keeping themselves healthy and well in everyday life.	Self-care
[126]	2023	Barlow et al. define "self-care" as the ability to manage symptoms, treatment, emotions, and lifestyle changes as part of living with a chronic condition.	Self-care
[32]	2024	Self-care refers to practices to maintain, recover, and improve one's health and wellbeing.	Self-care
[3]	2016	Intimate care has been defined as, "those care tasks associated with personal hygiene, bodily functions and bodily products, which demand direct or indirect contact with or exposure of the sexual parts of the body as defined culturally by the individual".	Intimate Care

Continued on next page

Table 3 – continued from previous page

Ref	Year	Quote	Define via
[176]	2023	In our research, we focus on the intimate care related to menstrual bleeding, which involves ways of handling and disposing of menstrual blood through absorbent products or objects that in other ways collect the blood, such as menstrual pads, menstrual cups, tampons and menstrual underwear.	Intimate Care
[6]	2022	In healthcare, care infrastructures do not only involve the "structures, functions, and resources required" to provide public and private healthcare services but also all the physical, social and structural arrangements of dynamic human and non-human elements that support and sustain care practices within and across different care settings.	Care Infrastructures
[125]	2023	For the purposes of this work, we consider virtual care to encompass the contexts in which healthcare providers leverage computer-mediated communication platforms to address the needs of remote patients.	Virtual Care
[140]	2016	For our purposes, any non-professional person who the patient identified as helping with some aspect of their care counts as a caregiver.	Caregiver
[130]	2017	Informal carers are those who provide care and support for another dependent person, often a family member, partner or spouse.	Informal Carers
[141]	2022	Home health aides are paid professional caregivers who provide home-based care for people with chronic conditions, who need long-term assistance, and/or who want to age in place. They help patients navigate the healthcare system and manage a wide range of chronic diseases by assisting with activities of daily living (e.g., bathing, grooming, grocery shopping, going to the doctor's office) and performing clinical monitoring (e.g., taking temperatures, recording mental/physical status).	Home health aides
[110]	2024	A key part of this economy is childcare work, which we define as the supervision and tending of children by someone other than the child's legal guardians.	Childcare Work
[91]	2017	Care work refers to work enacted in service of others, with common examples being child care, care provided for elderly parents, nursing, and teaching. Care work may be conducted either for pay or not for pay; however, its historical association with gendered divisions of labor, and the unpaid labor of family care, in particular, has contributed to it being rendered invisible in both research and broader society.	Care Work
[157]	2024	Beyond lateral care, the networks of care in the housecleaning space also incorporate people, technologies, spaces, and sociopolitical forces. We thus define "networks of care" as the full array of interactions between human and non-human actors which mutually infuence and enact care. Within networks of care, actors have the responsibilities of both care giver and receiver as care exchanges are multidirectional and afect all network actors.	Networks of Care
[157]	2024	Lateral care diverges from more traditional conceptualizations of care as it is focused around decentralized non-hierarchical networked systems of care, and - in the context of this study - enacted in digital spaces. These systems are characterized by shared identities and are bottom-up structures aimed at empowering their participants as they navigate diferential power hierarchies.	Lateral Care
[87]	2017	We use Fromm's definition of caring as the active concern for life and the growth of that which we love, and responsibility as the voluntary act of being ready to respond	Caring
[67]	2020	In the course of the study, we also discovered and explored an additional factor, care, which we define as instances in which people go beyond help by making sure the robot exits correctly or wishing it well upon its departure.	Care
[13]	2020	Care here, borrowing from the work of Lara Houston and Steven Jackson, refers to "[w]hat we do together to make the world a more liveable place".	Care
[164]	2022	Care refers to the feelings of attachment and responsibility towards ecosystems, which can include values, ideals, and moral attitudes towards the environment. Care can be connected to either values or principles stemming from particular relationships, or articulated by policies and social norms.	Care
[101]	2016	Our understanding of care leverages the 'beneficence' at the heart of Foucault's theoretical construct and builds on it with more recent feminist theories of care. [...] All these forms of care involve "a notion of doing and intervening" and have "strong affective and ethical connotations". Care is a practice and can be done well or badly.	Care

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Table 3 – continued from previous page

Ref	Year	Quote	Define via
[119]	2019	Care of the self, as we shall discuss in detail later, denotes "an exercise of the self on the self by which one attempts to develop and transform oneself, and to attain to a certain mode of being".	Care of the Self
[92]	2019	STS scholar Maria Puig de la Bellacasa describes care as "an everyday labour of maintenance that is also an ethical obligation: we must take care of things in order to remain responsible for their becomings".	Care
[167]	2019	The theoretical lens of care is also increasingly adopted in ICT discourse. Maria Puig de la Bellacasa posits care as affective enactment, an "ethicopolitical obligation", and a relational ontology; "to care about something, or for somebody, is inevitably to create relation". She invites us to attune to the care permeating everyday relations and make "care time" for what matters, be it in biological, sociotechnical, or scholarly contexts.	Care
[167]	2019	In this sense, care time offers "glimpses into a diversity of timelines that, despite being made invisible or marginalized in the dominant timescape, can challenge traditional notions of technoscientific innovation".	Care Time
[173]	2020	[...] builds upon the idea of cultivating care, a feminist approach to ethics that aims to engage students with a concept of responsibility to the other, interdependence, and attentiveness to vulnerability and inequality in the socio-technical systems we design.	Cultivating Care
[93]	2021	Caring about someone or something can be considered a basis of care through the discerning of caring needs. This might include caring about a particular child's happiness or more broadly income inequality, both of which imply an affective engagement with the wellbeing of a concern or responsibility.	Caring About
[93]	2021	Care labor can be considered the material practices that support caring. This is often discussed as the concrete tasks of care-giving in addressing caring needs.	Care Labour
[113]	2021	Care ethics can be understood as a feminist moral theory that privileges relationality, reciprocity, responsiveness, and particular (vs generalized) selves and others when confronting the implications of care labor and caring relations.	Care Ethics
[195]	2022	scholars have sought to advance frameworks of security as a set of moral obligations, in particular through the feminist ethic of care. Viewed through this lens, care is the establishment through social and technical infrastructures of ongoing care relations that prioritize maintenance of and continual affective attention to others' wellbeing. Care embraces broad consideration for the protection of people, centering collective responsibility and mutuality in the pursuit of fairer and more just sociotechnical systems.	Care Ethics
[145]	2022	It is worthwhile to briefly consider our conceptualisation of care within this paper, acknowledging it as a multi-level, relational construct which operates as a theoretical concept (e.g., a feminist ethics of care), and an organising force (how that ethics of care is operationalised). It has also been understood in terms of individual acts and activities of care (both formal and informal), directed both towards oneself and towards others	Care
[114]	2022	Care as a feminist practice is about attending to what, how, and when things get caring attention and come to matter and what, how, and when things don't. It is a relational, embodied, and ongoing practice which is necessarily particular.	Care
[2]	2023	We understand and interpret matters of care as an acute attention to: • entangled and situated relationships present in visualization collaborations; • neglected things as a source of necessary investigation and re-imagination; • and the power relations that shape the nature of these relationships before, during, and after collaborating. From this perspective, we understand research processes and artifacts as representations of relations and sites for enacting care.	Matters of Care

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Ref	Year	Quote	Define via
[75]	2023	As expressed by Puig de la Bellacasa, "representing matters of fact and sociotechnical assemblages as matters of care is to intervene in the articulation of ethically and politically demanding issues. The point is not only to expose or reveal invisible labours of care, but also to generate care". [...] "to represent matters of care is an aesthetic and political move in the way of re-presenting things that problematizes the neglect of caring relationalities in an assemblage. Here the meaning of care for knowledge producers might involve a modest attempt to share the burden of stratified worlds. This commitment is the political significance of representing matters of care."	Matters of Care
[59]	2024	Also known as care ethics (CE), ethics of care are a collection of approaches emphasizing interpersonal relationships, empathy, and caring for others in moral decision-making.	Care Ethics
[37, 93, 103, 112, 113, 120, 157, 162, 190, 207, 213, 216]	2015, 2021-2024	Refer to Tronto and Fisher's definition: "care as a species of activity that includes everything we do to maintain, contain, and repair our 'world' so that we can live in it as well as possible. That world includes our bodies, ourselves, and our environment".	Care